Contents

A Introduction **1**

B Minkowski space and space-like surfaces **1**
Space-like immersions. 2
Entire surfaces. 2
The Gauss map in $\mathbb{M}^{2,1}$ 4
$\mathbb{H}$ is a model for the Hyperbolic plane. 5

C The Gauss-Codazzi equations and the curvatures of S **6**
Recall on the second fundamental form 7

D Uniqueness for the PDE **7**
D.1 A generalized maximum principle 8
A boundedness result 11
D.2 The uniqueness theorem 15

A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

B Minkowski space and space-like surfaces

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as (x^1, x^2, x^3) or (x, y, t), choosing the notation according to convenience. We will also identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space V endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form g. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth nondegenerate $(0, 2)$-tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

Definition. Given a Lorentzian manifold (M, g), we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$
- light-like if $\langle v, v\rangle = 0$
- time-like if $\langle v, v\rangle < 0$

We will always denote with S a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. As usual, the immersion induces a nondegenerate bilinear form on S, the first fundamental form or pull-back pseudometric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we always forget the immersion ι).

Space-like immersions.
The object we will focus on are not arbitrary immersed surfaces but the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to the immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite hence Riemannian, or in other terms, they're surfaces S whose tangent bundle $\{(p, v)\}$ is made only of space-like vectors.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$.
We can show this with a simple argument:
First, we assumed that the pseudometric restricted to S,$\langle\cdot,\cdot\rangle|_S \geq 0$, is definite positive, which means that the tangent vectors of S at any point p belong to the set of space-like vectors:

$$T_p \subset \mathscr{C} := \left\{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\right\} = \{t^2 \leq x^2 + y^2\}.$$

This is just the outside-region of a double cone in $\mathbb{M}^{2,1}$, whose symmetry axis is $\left\{x^1 = 0, x^2 = 0\right\}$. A plane is entirely made of space-like vectors iff it's contained between the two branches of the cone.
Then, having assumed that S is an immersed surface, hence an immersed submanifold, it is locally the solution of an equation of the form $F(x, y, z) = 0$. It is well known that from this local expression we can express the normal vector to the surface at a point p: $N_p = \vec{\nabla} F(p)$.
Furthermore, we observe that N_p is orthogonal to the space-like plane T_pS and so it cannot be horizontal. The last two statements taken together give us $\partial_z F(p) = (N_p)^3 \neq 0$, and so we can apply the implicit function theorem and express the surface as the graph of $z = f(x, y)$ near the point p.
Where the local representation is valid, we can parametrize the surface as $(x, y, f(x, y))$, and derive the local frame $V_x = (1, 0, \partial_x f(x, y))$, $V_y = (1, 0, \partial_y f(x, y))$, for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x\rangle & \langle V_y, V_x\rangle \\ \langle V_x, V_y\rangle & \langle V_y, V_y\rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is obviously regular, the condition on g being a metric translates in having a strictly positive determinant,

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\vec{\nabla} f|^2,$$

which means $|\vec{\nabla} f|^2 < 1$.

Entire surfaces.
Given a surface S immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$.

We call the surface S is entire when the map Π is surjective. If the surface is globally a graph of a function φ, this condition is exactly stating that the function φ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

Proposition 1. *Let Ω be a domain in $\mathbb{C}$ (that is, an open connected set) and let $S : \Omega \to \mathbb{M}^{2,1}$ be a space-like immersion. Suppose that S is complete with respect to the induced Riemannian metric. Then S is an entire graph and Ω is simply connected.*

Before laying out the proof let's recall a couple of results from the theory of coverings.

Proposition 2. *Let M and N be two differentiable manifolds and $f : M \to N$ a smooth map*

- *If M is simply connected, N is connected and f is a covering map, then it is a homeomorphism.*
- *We say that the map f satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.*
- *If f is a local homeomorphism and satisfies the path lifting property, then it's a covering map.*

Proof. The idea of the proof is to show that $\Pi \circ S$ is a global diffeomrphism. Clearly this implies that Π restricted to S is a diffeomeorphism on $\mathbb{R}^2$ and so that S is an entire graph. From that we obtain also that Ω is simply connected since it's diffeomorphic to $\mathbb{R}^2$.

Let's observe that the target of $\Pi \circ S$ is just $\mathbb{R}^2$, which is simply connected, and the domain is Ω, which is connected since it's a domain. We already proved, by showing that S is locally a graph, that the map $\Pi \circ S$ is a local homeomorphism.
Having met these conditions, if we show that $\Pi \circ S$ satisfies the path lifting property, we conclude that it is a diffeomorphism.

Let's fix a point $z \in \Omega$ and a path $\gamma : [0,1] \to \mathbb{R}^2$ of bounded length, $\ell(\gamma) < \infty$, such that $\gamma(0) = (\Pi \circ S)(z)$. Our goal is to show that γ admits a lift to a path γ^* on Ω starting at z.
To this end, let's consider the set of times up to which such a lift exists,

$$\mathcal{J} := \{\tau \in [0,1] \mid \exists \gamma^* : [0,t] \to \Omega \text{ such that } (\Pi \circ S) \circ \gamma^* = \gamma \text{ and } \gamma^*(0) = z\},$$

and let's define $\bar{\tau} := \sup \mathcal{J}$.

If we prove that $\bar{\tau} = 1$, then γ admits a lift on the whole interval $[0,1]$ and we have finished. Then let's suppose by contradiction that $\bar{\tau} < 1$ and define $\gamma^* : [0, \bar{\tau}) \to \Omega$ the lift of γ up to $\bar{\tau}$. This curve exists by definition of $\bar{\tau}$.

Let's observe that $0 \in \mathcal{J}$ and that $\mathcal{J}$ is open. This is true because, being $\Pi \circ S$ a local homeomorphism, if a certain τ is in $\mathcal{J} \cap (0,1)$ then the whole interval $(\tau - \varepsilon, \tau + \varepsilon)$ for $\varepsilon > 0$ sufficiently small, belongs to it.

Finally, we need to prove that the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(t)$ exists. If this is true we can then extend γ^* up to $\bar{\tau}$, which means that $\bar{\tau} \in \mathcal{J}$. Since $\mathcal{J}$ is open, there exists an element $\bar{\tau} + \varepsilon \in \mathcal{J},\ \varepsilon > 0$. This contradicts the definition of $\bar{\tau}$ and proves $\bar{\tau} = 1$ since the alternative is absurd.

To prove that $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists we just need a couple of observations:
First we note that the map Π dilates distances, as

$$\begin{aligned}\langle v, v \rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v \rangle_{\mathbb{R}^2}\,.\end{aligned}$$

From this it's clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
Now, if we fix any sequence $\tau_k \to \bar{\tau}$ and observe that as γ is differentiable, it's derivative is bounded, so $\ell(\gamma|_{[\tau_k, \tau_m]}) \leq C|\tau_k - \tau_m|$. By observing that

$$\begin{aligned}d(\gamma^*(\tau_k), \gamma^*(\tau_m)) &\leq \ell(\gamma^*|_{[\tau_k, \tau_m]}) \\ &\leq \ell(\gamma|_{[\tau_k, \tau_m]}) \\ &\leq C|\tau_k - \tau_m|\end{aligned}$$

we deduce that also $(\gamma^*(\tau_k))_k$ is a Cauchy sequence, hence converging since S is complete and Ω is to be intended with the induced metric.
By the arbitrariness of the sequence $(\tau_k)_k$ we deduce the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists. $\square$

In this section we have essentially proved that a complete space-like surface, immersed in $\mathbb{M}^{2,1}$ is actually the graph of a map with $|\vec{\nabla} f|^2 < 1$. It's easy to convince ourselves that the inverse result also holds and so that this is actually a characterization.

The Gauss map in $\mathbb{M}^{2,1}$
The surface S is an immersed submanifold of codimension 1. As we have a nondegenerate form on the ambient space, g, once fixed a $p \in S$, we can write the following decomposition, which varies smoothly as we change the point:

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS$$

The codimension of the submanifold M is equal to the dimension of $(T_pS)^\perp$, that is, is exaclty one. In fact this space is generated only by the normal vector to S, N_p.
We can easily show that N must be a time-like vector, due to the fact that T_pS is made of space-like vectors.

Let's fix $e_1, e_2 \in T_pM$ two independent vectors. Since N belongs to $(T_pM)^\perp$ we have $\langle e_1, N \rangle = \langle e_2, N \rangle = 0$. This implies that when we look at the represen-

tative matrix of g_p written in the basis (N, e_1, e_2), we have

$$[g_p] = \begin{pmatrix} \langle n, n\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle n, n\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form g restricted to the subspace T_pM. We conclude that $\langle n, n\rangle$ is negative just by observing that $\det[g_p] < 0$ as g is a pseudometric, while $\det[(g|_S)_p] > 0$ since g becomes a metric when restricted to S.

We can now repeat the classical construction of the Gauss map in this pseusoriemannian context.
The idea is just to take a unit vector in $(T_pM)^\perp$. On this space the norm of the vectors is negative so we will need to normalize them to -1 instead of 1.
So let's fix $w \in (T_pM)^\perp$ and scale it by a factor k such that $\langle kw, kw\rangle = -1$. If we solve this equation we obtain two distinct solutions, that send w to one of the two branches of $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} |\, \langle v, v\rangle = -1\}$, which means that a choice is needed. Here we consistently choose the normalized vector to to be in the space

$$\mathbb{H} = \mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} |\, \langle v, v\rangle = -1, v^3 > 0\}.$$

This way we have constructed a map

$$u : S \to \mathbb{H},$$

the Gauss map of the surface S. Of course the only difference from the classical theory of surfaces is the codomain of the map u, where it is the sphere $\mathbb{S}^2$. Like in the classical case, the choice of the normal vector specifies an orientation for the surface, which implies that the Gauss map is gloabally defined only when the surface is orientable. In our case, this condition is trivially met since we're dealing with a graph surface.

$\mathbb{H}$ is a model for the Hyperbolic plane.
The space $\mathbb{H}$ that we just defined is clearly a surface immersed in $\mathbb{M}^{2,1}$, but there is more. First of all, we observe that, by manipulating the definition of $\mathbb{H}$ we can write it as the graph of the function $\varphi(x, y) = \sqrt{x^2 + y^2 + 1}$. The gradient of such function is $\nabla\varphi = \frac{1}{\sqrt{x^2+y^2+1}}(x, y)$, from which it's clear that $|\nabla\varphi| < 1$. By applying the result we proved before, we obtain that the bilinear form that $\mathbb{H}$ inherited from the immersion in $\mathbb{M}^{2,1}$ is a Riemannian metric.
In fact this space is one of the many models of the hyperbolic plane. Amongst these models there is one which might be more useful than others, the Poincaré disk. As a manifold, the Poincaré disk is simply the unit ball in $\mathbb{R}^2$, $\mathbb{D} = \{\xi \in \mathbb{R}^2 \mid |\xi| \leq 1\}$, while we can construct the metric by exhibiting a diffeomorphism between $\mathbb{D}$ and $\mathbb{H}$ pulling back the metric.
A possible diffeomorphism is the map p, whose inverse is the map q, as can be verified by direct computation:

$$\begin{aligned} p\colon \mathbb{H} &\longrightarrow \mathbb{D} \\ (x, y, t) &\longmapsto \frac{1}{1+t}(x, y) \end{aligned} \qquad \begin{aligned} q\colon \mathbb{D} &\longrightarrow \mathbb{H} \\ \xi &\longmapsto \frac{1}{1-|\xi|^2}(2\xi^1, 2\xi^2, 1 + |\xi|^2). \end{aligned}$$

It's clear that both functions are smooth, since $|\xi| < 1$ and $t > 1$.
We would like to find the expression of $q^*(g_{\mathbb{H}})$.To simplify the calculations we might observe that the metric of $\mathbb{H}$ inherited by $\mathbb{M}^{2,1}$ is invariant under a rotation on the t axis, and that such a rotation corresponds under p to a rotation of the disk. This means that the metric $q^*(g_H)$ will be invariant under rotation.
It is then sufficient to fix a certain $\xi \in \mathbb{D}$ of the form $\xi = (a, 0)$ and calculate $q^*(g_{\mathbb{H}})_\xi$; this will specify the whole metric tensor. By direct computation we obtain

$$[dq_\xi] = \frac{2}{(1-|\xi|^2)^2} \begin{pmatrix} 1+(\xi^1)^2-(\xi^2)^2 & 2\xi^1\xi^2 \\ 2\xi^1\xi^2 & 1-(\xi^1)^2+(\xi^2)^2 \\ 2\xi^1 & 2\xi^2 \end{pmatrix}.$$

Which, evaluated at $\xi = (a, 0)$, yields

$$[dq_{(a,0)}] = \frac{2}{1-a^2} \begin{pmatrix} \frac{1+a^2}{1-a^2} & 0 \\ 0 & 1 \\ \frac{2a}{1-a^2} & 0 \end{pmatrix} =: \frac{2}{1-a^2} \left(v_1 | v_2\right).$$

We observe that, with respect to the metric of $\mathbb{M}^{2,1}$, it holds

$$\langle v_1, v_1 \rangle = 1, \qquad \langle v_2, v_2 \rangle = 1, \qquad \langle v_1, v_2 \rangle = 0.$$

Which means that the representative matrix of $q^*(g_{\mathbb{H}})$ is a multiple of the identity, $[q^*(g_{\mathbb{H}})_{(0,a)}] = \frac{2}{1-a^2} I$. By the argument on rotational invariance, we conclude that for every $\xi \in \mathbb{D}$,

$$q^*(g_{\mathbb{H}})_\xi = \left(\frac{2}{1-|\xi|^2}\right)^2 g^E_\xi.$$

The disk $\mathbb{D}$ with this metric is called the Poincaré model of the Hyperbolic plane. A nice property of this model is to be what is called a "conformal" model of the hyperbolic plane, namely an immersed manifold in a Euclidean space whose metric is a multiple of the standard one.

What we discussed in this paragraph lets us identify $\mathbb{H}$ and $\mathbb{D}$. From now on we will use $\mathbb{D}$. For example, we will always use this space as the codomain of the Gauss map,

$$u : S \to \mathbb{D}.$$

C The Gauss-Codazzi equations and the curvatures of S

In this section we will take a look at a pair of equations that encode much of the information of an immersion, the so called Gauss-Codazzi equations.

Recall on the second fundamental form
Let's give a recall of the second fundamental form, which as it's well known, encodes how the surface curves in the ambient space.
Let's suppose to have two Riemannian manifolds and an isometric immersion between them $f : M^{(n)} \to \overline{M}^{(n+k)}$, for which we will identify M as a subset of $\overline{M}$.
It is clear that any smooth vector field on M can be extended to a smooth vector field on $\overline{M}$ and that the ambient metric induces the following splitting

$$T_p\overline{M} = T_pM \oplus (T_pM)^{\perp},$$

which is smooth, in the sense that the maps $(p, v) \mapsto (p, v^T)$ and $(p, v) \mapsto (p, v^N)$ are smooth transformations of TM.

Let's fix a connection $\overline{\nabla}$ on $\overline{M}$ and denote with ∇ the induced connection on M, which is constructed by taking two vector fields on M, X, Y, extending them to $\overline{M}$ obtaining $\overline{X}$ and $\overline{Y}$, and finally by projecting on the tangent space of M, $\nabla_X Y = (\overline{\nabla}_{\overline{X}}\overline{Y})^T$. We then define the bilinear form

$$B(X, Y) = \overline{\nabla}_{\overline{X}}\overline{Y} - \nabla_X Y.$$

It is clear that $B(X, Y)_p \perp T_pM$ everywhere. This means that given a small open set $U \subset M$, we can intend B as a function between the space of vector fields on U tangent to M, $\mathcal{X}(U)$ and those normal to M, $\mathcal{X}(U)^{\perp}$.

$$B : \mathcal{X}(U) \times \mathcal{X}(U) \to \mathcal{X}(U)^{\perp}$$

It can be seen also that B is $C^{\infty}(M)$-linear and symmetric, or in other words that it's a symmetric tensor. This means in particular that we can evaluate $B(x, y)$ at specific vectors $x, y \in T_pM$ and do not need the whole vector field.
We have all we need to define the second fundamental form. Given $p \in M$ and a vector $\eta \in T_pM^{\perp}$, the mapping

$$\begin{aligned} H_\eta : T_pM \times T_pM &\longrightarrow \mathbb{R} \\ (x, y) &\longmapsto \langle B(x, y), \eta \rangle \end{aligned}$$

is the second fundamental form of the immersion f at the point p along the normal vector η.

D Uniqueness for the PDE

So far we have reduced our classification problem to proving existence and uniqueness for the following differential problem.

Given h a smooth function on the Poincaré disk $\mathbb{D}$, find a smooth function

$$w : \mathbb{D} \to \mathbb{R}$$

such that

$$\Delta_p w = e^{2w} - h^2\, e^{-2w} - 1,$$

where Δ_p is the Laplacian with respect to the metric ds_p^2 and such that

$$e^{2w} ds_p^2$$

is a complete metric on D, with

$$e^{2w} - h^2 e^{-2w} > 0$$

on $\mathbb{D}$.
rimuoverò questo box quando ho scritto le sezioni precedenti

We will need the following generalized maximum principle, due to Cheng and Yau [1]. ciao

D.1 A generalized maximum principle

Let's lay out the setup. Let's suppose to have a complete Riemannian manifold of dimension n, $M^{(n)}$ and to have fixed an arbitrary point p. We suppose that the manifold $M^{(n)}$ has nonpositive sectional curvature. This hypothesis is not really necessary for what is concerned in this paragraph, but it simplifies greatly the notation and is verified in the applied problem we are facing. Let $\sigma : [0, l] \to M$ be a unit speed geodesic joining x to p, whose velocity is $\sigma'(t)$ and where the Ricci curvature is given by $\mathrm{Ric}(\sigma'(t))$.
Let's define

$$K_\sigma(x) = \min_{0 \le k \le l} \left[\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \,\mathrm{Ric}(\sigma'(t)) dt \right].$$

Since our manifold has, by hypothesis, a nonpositive sectional curvature, by applying the Cartan-hadamard theorem we can assume that the cut locus of p is empty, and so that σ is the shortest geodesic between x and p.
Let's define $r(x) := d(x, p)$ the distance in M from p. By the means of a simple computation, we can observe that when $r(x) \ge 1$ and if the Ricci curvature is bounded from below, then the quantity $K(x)$ is bounded from above by a constant.sistemare

Theorem D.1. *Let M be a complete Riemannian manifold of nonpositive sectional curvature and $f \in C^2(M)$ a function bounded from above. Fix $p \in M$, and let $r(x) = d(p, x)$. Then there exists a sequence $(x_k)_k \subset M$ such that*

$$f(x_k) \to \sup_M f, \qquad |\nabla f|(x_k) \le \frac{1}{k} \to 0, \qquad \limsup_{k\to\infty} \Delta f(x_k) \le 0.$$

If moreover the Ricci curvature is bounded from below, we can choose the sequence $(x_k)_k$ so that the following estimates also hold,

$$\Delta f(x_k) \le \frac{C}{k}$$

for some constant $C > 0$ independent from k.

{th:maxprin}

Theorem D.2. *Let M be a complete Riemannian manifold of nonpositive sectional curvature. Let f any C^2 function bounded from above on M. Then for any $p \in M$ there is a sequence of points $(x_k)_k \subset M$ such that*

a.
$$\lim_{k\to\infty} f(x_k) = \sup f$$

b.
$$|\nabla f|(x_k) = \frac{2(f(x_k) - f(p) + 1)r(x_k)}{k(r^2(x_k) + 2)\log(r^2(x_k) + 2)}$$

c.
$$\Delta f(x_m) = \frac{2\big(f(x_m) - f(p) + 1\big)\big(r(x_m)K(x_m) + 1\big)}{m\big(r(x_m)^2 + 2\big)^2 \log\big(r(x_m)^2 + 2\big)} \tag{1}$$
$$+ \frac{4\big(f(x_m) - f(p) + 1\big)r(x_m)^2}{m\big(r(x_m)^2 + 2\big)^2 \log\big(r(x_m)^2 + 2\big)}. \tag{2}$$

interpretazione di questo big mess
aggiungere statement teorema Caratan-Hadamard e teorema "$r(x) = d(p,x)^2$ differenziabile in x se x non è nel cut locus di p".
aggiungere osservazione per cui una funzione ottiene il suo massimo in x allora ha laplaciano non positivo in x.

Proof. Let's define the auxiliary functions $u(x) = (f(x) - f(p) + 1)$ and $L(x) = \log(r^2(x) + 2)$. Then let's fix a positive integer k and define the function g_k as

$$g(x) = \frac{f(x) - f(p) + 1}{(\log(r^2(x) + 2)^{\frac{1}{k}}} = \frac{u(x)}{L(x)^{\frac{1}{k}}} \quad .$$

Let's observe that $g(p) = \frac{1}{\log(2)^{\frac{1}{k}}} > 0$ and that $\lim_{r(x)\to\infty} g(x) = 0$. Since g is continuous, this means that it attains its maximum at a certain point x_k. why
As we already mentioned, the Cartan-Hadamard M theorem grants that x_k is not in the cut-locus of p, and so that the function $d(p,x)^2 = r(x)^2$ is differentiable in x.
Let's calculate the gradient of g,

$$\nabla g(x) = \frac{\nabla f(x)}{\log(r(x)^2 + 2)^{1/k}} - \frac{2(f(x) - f(p) + 1)r(x)}{k(r(x)^2 + 2)\log(r(x)^2 + 2)^{1+1/k}}\nabla r(x) \quad .$$

x_k is the maximum of g_k, which means $\nabla g(x_k) = 0$, from which we get, noting also that r satisfies $|\nabla r| = 1$,

$$|\nabla f|(x_k) = \frac{2u(x_k)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \quad .$$

We then observe, since f is bounded from above, that $u(x) = f(x) - f(p) + 1 \leq \sup f - f(p) + 1 \leq 1$; also, the map $\frac{2x}{(x^2+2)\log(x^2+2)} \leq 2$, so that we get the estimate

$$|\nabla f|(x_k) \leq \frac{2}{k} \quad .$$

Let's now focus on the estimate on the Laplacian. Again by direct computation we obtain

$$\Delta g = \frac{\Delta f}{L^{1/k}} - \frac{4r}{k(r^2+2)L^{1+1/k}}\langle \nabla f, \nabla r\rangle - \frac{2u}{k(r^2+2)L^{1+1/k}} - \frac{2ur\,\Delta r}{k(r^2+2)L^{1+1/k}} + \frac{4ur^2}{k(r^2+2)^2L^{1+1/k}} + \frac{4(k+1)ur^2}{k^2(r^2+2)^2L^{2+1/k}}.$$

By observing that $\Delta g(x_k) \leq 0$ since x_k is a maximum, substituting the expression for Δg, the fact that $\nabla g(x_k) = 0$ and observing that controllare

$$\langle \nabla f, \nabla r\rangle(x_k) = \frac{2\,u(x_k)\,r(x_k)}{k\big(r^2(x_k)+2\big)\log\big(r^2(x_k)+2\big)} \quad ,$$

we get

$$\Delta f(x_k) \leq \frac{2u(x_k)\big(1+r(x_k)\Delta r(x_k)\big)}{k\big(r^2(x_k)+2\big)\log\big(r^2(x_k)+2\big)} - \frac{4u(x_k)r^2(x_k)}{k\big(r^2(x_k)+2\big)^2\log\big(r^2(x_k)+2\big)} - \frac{4(k-1)u(x_k)r^2(x_k)}{k^2\big(r^2(x_k)+2\big)^2\log^2\big(r^2(x_k)+2\big)} \quad .$$

And in particular

$$\Delta f(x_k) \leq \frac{2u(x_k)\big(1+r(x_k)\Delta r(x_k)\big)}{k\,(r^2(x_k)+2)\,\log(r^2(x_k)+2)} \quad .$$

We now apply the Cheng-Yau estimate

$$\Delta r(x) \leq K(x),$$

where $K(x)$ is the quantity defined along minimizing geodesics. If the Ricci curvature is bounded from below, then $K(x)$ is bounded above for $r(x) \geq 1$. Therefore, if $u \leq 1$, there exists a constant $C > 0$ such that

$$\Delta f(x_k) \leq \frac{C}{k}.$$

assolutamente da riscrivere

We now show directly that

$$f(x_k) \to \sup_M f.$$

Recall that x_k is chosen so that

$$g_k(x_k) = \sup_M g_k,$$

where

$$g_k(x) = \frac{f(x) - f(p) + 1}{L(x)^{1/k}}, \qquad L(x) := \log\big(r(x)^2 + 2\big).$$

Fix $x \in M$. Since x_k maximizes g_k, we have

$$g_k(x_k) \geq g_k(x) = \frac{f(x) - f(p) + 1}{L(x)^{1/k}}.$$

Now x is fixed, hence $L(x)$ is fixed, and therefore

$$L(x)^{1/k} \to 1 \qquad \text{as } k \to \infty.$$

It follows that

$$\liminf_{k\to\infty} g_k(x_k) \geq f(x) - f(p) + 1.$$

Since this holds for every $x \in M$, we obtain

$$\liminf_{k\to\infty} g_k(x_k) \geq \sup_M f - f(p) + 1.$$

On the other hand, since $f \leq \sup_M f$ and $L(x_k)^{1/k} \geq 1$, we have

$$g_k(x_k) = \frac{f(x_k) - f(p) + 1}{L(x_k)^{1/k}} \leq \sup_M f - f(p) + 1.$$

Therefore

$$\limsup_{k\to\infty} g_k(x_k) \leq \sup_M f - f(p) + 1.$$

Combining the two inequalities yields

$$g_k(x_k) \to \sup_M f - f(p) + 1.$$

Finally, since $L(x_k)^{1/k} \geq 1$, one has

$$g_k(x_k) = \frac{f(x_k) - f(p) + 1}{L(x_k)^{1/k}} \leq f(x_k) - f(p) + 1 \leq \sup_M f - f(p) + 1.$$

Passing to the limit and using the convergence of $g_k(x_k)$, we get

$$\sup_M f - f(p) + 1 \leq \liminf_{k\to\infty}\big(f(x_k) - f(p) + 1\big) \leq \limsup_{k\to\infty}\big(f(x_k) - f(p) + 1\big) \leq \sup_M f - f(p) + 1.$$

Hence

$$f(x_k) \to \sup_M f.$$

□

A boundedness result

We can use this theorem also to prove a boundedness result for a certain differential inequality. We will make some observation and state the theorem at the end of the section.

Let's assume that M is a Riemannian Manifold with is Ricci curvature bounded from below. Let u be a C^2 function such that

$$\Delta u \geq f \quad , \tag{3}$$ {eq:boundprin2}

where we will state the conditions that f must satisfy later.

Let's suppose to have a positive function $F \in C^2(\mathbb{R})$ that also we will define later. Since F is positive, the function $-1/F$ is bounded from above. So we can apply theorem D.2 and find a sequence $(x_k)_k$ such that

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = -\sup\left(-\frac{1}{F(u)}\right) = \inf \frac{1}{F(u)} \tag{4}$$ {eq:boundprin7}

and along with the estimates on the gradient and the Laplacian apply,

$$\frac{|F'(u)||\nabla u|(x_k)}{F^2(u)} \leq \frac{1}{k}, \tag{5}$$ {eq:boundprin3}

$$-\frac{F''(u)\,|\nabla u|^2}{F^2}(x_k) - \frac{F'(u)\,\Delta u}{F^2}(x_k) + \frac{(F'(u))^2\,|\nabla u|^2}{F^3}(x_k) \geq -\frac{1}{k} \quad .\textit{align me} \tag{6}$$

By multiplying the last equation by $F'(u)/(F(u)^2|F''(u)|$ we get,

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} - \frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} + \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \geq -\frac{F'(u)^2}{kF(u)^2|F''(u)|} \quad . \tag{7}$$ {eq:boundprin1}

Now we make some estimates on the LHS in (7):

$$\frac{F'(u)^3}{F^4}\frac{\nabla u}{|F(u)|} \geq \frac{F'(u)^3}{F^4}\frac{f(u)}{|F(u)|} \quad \text{since } \Delta u \geq f, \text{ (3)}$$

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq \left|-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| = \left|\frac{(F')^2|\nabla u|^2}{F^4}\right| \leq \frac{1}{k^2} \qquad \text{apply [(5)]} \quad .$$

also,

$$\frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq 2\left|\frac{F'(u)}{F(u)|F''(u)|}\right|\left|\frac{F'(u)^2|\nabla u|^2}{F^4}\right| \leq \frac{2}{k^2}\left|\frac{F'(u)}{F(u)|F''(u)|}\right| \quad \text{for (5)} \quad .$$

So that we get from (5),

$$\frac{1}{k^2} - \frac{F'(u)^3 f(u)}{F^4|F(u)|} + \frac{2}{k^2}\left|\frac{F'(u)}{F(u)|F''(u)|}\right| \geq -\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|} \quad . \tag{8}$$ {eq:boundprin6}

If we suppose that our function F satisfies

$$\limsup_{t\to\infty} \frac{|F'(t)|^2}{F(t)\,|F''(t)|} < \infty \tag{9}$$ {eq:boundprin4}

and

$$\liminf_{t\to\infty} \frac{F'(t)^3 f(t)}{F(t)^4\,|F''(t)|} > 0 \quad , \tag{10}$$ {eq:boundprin5}

we can prove then that $\limsup_k u(x_k) < \infty$.

Let's suppose by contradiction that $\limsup u(x_k) = \infty$, then we can take $\limsup$ in (8)

$$\limsup_k k \to \infty \left(\frac{1}{k^2} - \frac{F'(u)^3 f(u)}{F^4|F(u)|} + \frac{2}{k^2}\left|\frac{F'(u)}{F(u)|F''(u)|}\right|\right) \geq \limsup_{k\to\infty}\left(-\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\right)$$

$$\limsup_{k\to\infty}(\frac{1}{k^2}) - \liminf_{k\to\infty}\left(\frac{F'(u)^3 f(u)}{F^4|F(u)|}\right) + 2\lim_{k\to\infty}\left(\frac{1}{k^2}\right)\limsup_{k\to\infty}\left(\frac{F'(u)^3 f(u)}{F^4|F(u)|}\right)$$
$$\geq \limsup_{k\to\infty}\left(-\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\right) \quad .$$

since we assumed (9), this is equivalent to

$$\begin{aligned}\liminf_{k\to\infty}\left(\frac{F'(u)^3 f(u)}{F^4|F(u)|}\right) &\leq -\limsup_{k\to\infty}\left(-\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\right)\\ &= -\lim_{k\to\infty}\left(-\frac{1}{k}\right)\limsup_{k\to\infty}\left(\frac{F'(u)^2}{F(u)^2|F''(u)|}\right)\\ &= 0 \quad .\end{aligned}$$

Again having used (9). This contradicts the assumption (10). We have proven that $\limsup_k u(x_k) < \infty$, so up to taking a subsequence, we can assume that $u(x_k)$ admits a limit.

Now it's a good point to prove that u is bounded, provided the existence of the function F, which we need to be strictly increasing so that $1/F$ is strictly decreasing.
First we recall that if $h : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then

$$u(x_k) \to \sup(u) \Leftrightarrow h \circ u(x_k) \to \inf(h \circ u) \quad .$$

In (4) we found that the sequence $(x_k)_k$ satisfies

$$\begin{aligned}\lim_{k\to\infty}\frac{1}{F(u(x_k))} &= \inf\frac{1}{F(u)}\\ &= \frac{1}{\sup F(u)} \quad .\end{aligned}$$

hence, knowing that $F > 0$, $\lim F(u(x_k)) = \sup F(u)$. So, knowing that $\limsup_k u(x_k) \leq C$,

$$\begin{aligned}\sup(F(u)) &= \lim F(u(x_k))\\ \Rightarrow \sup(u) &= \lim_k u(x_k) \leq 0 \quad .\end{aligned}$$

We have moved our problem to finding a function F that is

- Positive
- Strictly increasing
- Satisfies (9) and (10) .

Let's search such a function F of the form

$$F(s) = \int_a^s \left(\int_a^t g(\tau)d\tau\right)^{-\frac{1}{2}} dt + 1 \quad ,$$

where g is a positive continuous function defined on $[a, \infty)$ with $\int_b^\infty \left(\int_a^t g(\tau)d\tau\right)^{-1/2} dt < \infty$. Then we extend F on the whole line $\mathbb{R}$ so that it's a strictly increasing C^2

function.
We need to compute a couple of quantities,

$$F''(s) = -\frac{1}{2}\left(\int_a^s g(\tau)d\tau\right)^{-\frac{3}{2}} g(s) \quad .$$

$$\frac{|F'(s)|^2}{F(t)|F''(t)|} = 2\frac{\left(\int_a^t g(\tau)d\tau\right)^{\frac{1}{2}}}{F(t)g(t)} \quad .$$

so the condition (9) is equivalent to

$$\limsup_{t\to\infty} \frac{\left(\int_a^t g(\tau)\, d\tau\right)^{1/2}}{F(t)\, g(t)} < \infty \quad .$$

$$\frac{F'(t)^3 f(t)}{F(t)^4|F''(t)|} = 2\frac{f(t)}{F(t)^4 g(t)} \quad .$$

Then also the condition (10) is equivalent to

$$\liminf_{t\to\infty} \frac{f(t)}{F(t)^4 g(t)} > 0.$$

Finally we observe that since g is positive, if $t < s$,

$$\left(\int_a^t g(\tau)d\tau\right)^{-1/2} \geq \left(\int_a^s g(\tau)d\tau\right)^{-1/2} \quad ,$$

so that

$$\begin{aligned} F(t) - 1 &= \int_a^s \left(\int_a^t g(\tau)d\tau\right)^{-\frac{1}{2}} ds \\ &\geq \int_a^t \left(\int_a^t g(\tau)d\tau\right)^{-\frac{1}{2}} dt \\ &= \left(\int_a^t g(\tau)d\tau\right)^{-1/2} \int_b^t 1ds \\ &= (t-b)\left(\int_a^t g(\tau)d\tau\right) \quad . \end{aligned}$$

So

$$F(t) \geq (t-b)\left(\int_a^t g(\tau)d\tau\right)^{-1/2} \quad .$$

Which means we can further rewrite the condition (10)

$$\frac{\left(\int_a^t g(\tau)\,d\tau\right)^{1/2}}{F(t)\,g(t)} \leq \frac{\left(\int_a^t g(\tau)\,d\tau\right)^{1/2}}{(t-b)\left(\int_a^t g(\tau)d\tau\right)^{-1/2} g(t)}$$
$$= \frac{\int_a^t g(\tau)d\tau}{(t-b)g(t)}$$
$$\sim \frac{\int_a^t g(\tau)d\tau}{tg(t)} \quad .$$

We have proven the following theorem

{th:boundprin}

Theorem D.3. *Let M be a Riemannian manifold whose Ricci curvature is bounded from below such that*

$$\Delta u \geq f(u) \quad .$$

A sufficient condition for u being bounded from above on M is that there is a continuous positive function g, defined on an interval $[a,\infty)$ such that

- $\int_b^\infty \left(\int_a^t g(\tau)d\tau\right)^{-1/2} dt < \infty$ *for some* $b \geq a$,
- $\limsup_{t\to\infty} \frac{\int_a^t g(\tau)d\tau}{tg(t)} < \infty$
- $\liminf_{t\to\infty} \frac{f(t)}{g(t)} > 0$.

D.2 The uniqueness theorem

In order to prove uniqueness, we pass to an auxiliary formulation of the problem. We first prove uniqueness in this new setting, and then show that this implies uniqueness for the original equation.
Recall that we are looking for a function w such that the metric g can be written as

$$g = e^{2w} ds_p^2,$$

where ds_p^2 is the Poincaré metric on $\mathbb{D}$. Since ds_p^2 is itself conformal to the Euclidean metric $|dz|^2$, with conformal factor

$$\frac{4}{(1-|z|^2)^2},$$

it follows that g is also conformal to $|dz|^2$. In fact, we can write

$$g = \frac{4e^{2w}}{(1-|z|^2)^2}|dz|^2.$$

If we now express the conformal factor in the form $e^{2\tilde{w}}$, we find a transformation between w and $\tilde{w}$,

$$\tilde{w} = w + \log\left(\frac{2}{1-|z|^2}\right).$$

This lets us write the metric g as

$$g = e^{2\tilde{w}}|dz|^2.$$

Therefore w and $\tilde{w}$ describe the same metric, the former with respect to the Poincaré background metric and the latter with respect to the Euclidean one. We perform this transformation as it will be convenient to study the corresponding equation in the Euclidean setting first, where the relevant operator is the standard Laplacian Δ. Once we have proved uniqueness there, we will translate the result back to the original formulation.

Theorem D.4. *Let h be a continuous function defined on a domain Ω in $\mathbb{C}$. Let $\tilde{w}$ be a solution for [where do we use the inequality with zero?]*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{11}$$ {eq:unithm}

in Ω such that $e^{2\tilde{w}}$ is a complete metric on Ω. Then any other solution v of the equation satisfies $v \leq \tilde{w}$.

Proof. Let us define $\eta = v - \tilde{w}$. We need to prove that $\eta \leq 0$.
We have

$$\begin{aligned}\Delta\eta &= \Delta v - \Delta\tilde{w} \\ &\overset{[(11)]}{=} e^{2v} - h^2 e^{-2v} - e^{2\tilde{w}} + h^2 e^{-2\tilde{w}} \quad .\end{aligned} \tag{12}$$ {eq:unithm2}

Now let's make some calculations

$$e^{2v} = e^{2(\eta+\tilde{w})} = e^{2\eta}e^{2\tilde{w}} \quad .$$

Let's define the quantity $|\mu|^2 = h^2 e^{-4\tilde{w}}$, we have then

$$\begin{aligned} h^2 e^{-2\tilde{w}} &= h^2 e^{-2\tilde{w}} e^{-2\eta} = |\mu|^2 e^{2\tilde{w}} e^{-2\eta} \\ h^2 e^{-2\tilde{w}} &= \left(h^2 e^{-4\tilde{w}}\right) e^{2\tilde{w}} = |\mu|^2 e^{2\tilde{w}} \quad . \end{aligned}$$

So, by substituting what we just found in (12), we get

$$\begin{aligned}\Delta\eta &= e^{2\tilde{w}}e^{2\eta} - |\mu|^2 e^{2\tilde{w}} e^{-2\eta} - e^{2\tilde{w}} + |\mu|^2 e^{2\tilde{w}} \\ &= e^{2\tilde{w}}(e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \quad .\end{aligned} \tag{13}$$ {eq:unithm3}

The Laplacian with respect to the metric $g = e^{2\tilde{w}}|dz|^2$ is simply $e^{-2\tilde{w}}\Delta$. So in (13) we obtain

$$\Delta_g \eta = e^{-2\tilde{w}}\Delta\eta = e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \quad .$$

Let's now observe that $|\mu|^2 \leq 1$, since we know from (11)

$$\begin{aligned} e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} &\geq 0 \\ e^{2\tilde{w}}(1 - h^2 e^{-4\tilde{w}}) &\geq 0 \\ e^{2\tilde{w}}(1 - |\mu|^2) &\geq 0 \quad , \end{aligned}$$

so that $|\mu| \leq 1$.
Now, continuing in (13),

$$\begin{aligned}\Delta_g \eta &= e^{2\tilde{w}}(e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \\ &\geq e^{2\eta} - e^{-2\eta} - 1 + |\mu|^2 \\ &\geq e^{2\eta} - e^{-2\eta} - 1 \\ &=: f(\eta) \quad .\end{aligned}$$

This inequality is exactly what is found in theorem D.3. If we show the other hypotheses, we obtain that η is bounded. First, a direct computation shows that $g(t) = t^2 \chi_{[0,\infty)}$, where χ is the indicator function of the interval, satisfies the three required conditions.
Also, the metric g is assumed to be complete, so we only need to check that the Gaussian curvature (which is the same as the Ricci curvature since we're dealing with a surface) is bounded from below:questa parte è da sistemare

.

Now, since η is bounded from above, the Gaussian curvature of g is negative, and it is clearly smooth, we can apply also theorem D.2, obtaining a sequence $(z_k)_k \subset \Omega$ such that

$$\lim_{k\to\infty} \eta(z_k) = \bar{\eta} = \sup \eta, \quad \text{and} \quad \limsup_{k\to\infty} \Delta_g \eta(z_k) \leq 0 \quad .$$

We know that $|\mu|^2 \leq 1$ is continuous and bounded, so we can assume that the sequence verifies

$$\lim_{k\to\infty} |\mu|^2(z_k) = a \leq 1 \quad .$$

Therefore, by taking the limits in (13), we obtain

$$\begin{aligned}0 \geq \limsup_{k\to\infty} \Delta_g \eta(z_k) &= \lim_{k\to\infty} \left(e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2\right) \\ &= e^{2\bar{\eta}} - a e^{-2\bar{\eta}} - 1 + a \quad . \end{aligned} \tag{14}$$ {eq:unithm4}

A simple computation shows that the function $\varphi(s) = e^{2s} - a e^{-2s} - 1 + a$ is increasing and such that $\varphi(0) = 0$. So from (14) we get that $\sup \eta\bar{\eta} \leq 0$, which means $\eta \leq 0$ everywhere. □

Corollary D.1. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$ and let $\tilde{w}_1, \tilde{w}_2$ two solutions of*

$$\Delta \tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{15}$$

in Ω such that $e^{2\tilde{w}_i}$ are complete metrics on Ω. Then $\tilde{w}_1 \equiv \tilde{w}_2$.

Corollary D.2. *There is at most one solution of the equation ??? such that the associated metric is complete.*

References

[1] Shiu Yuen Cheng and Shing-Tung Yau. "Differential equations on Riemannian manifolds and their geometric applications". In: *Communications on Pure and Applied Mathematics* 28.3 (1975), pp. 333–354.

[2] Shing-Tung Yau. "Harmonic functions on complete Riemannian manifolds". In: *Communications on Pure and Applied Mathematics* 28.2 (1975), pp. 201–228.